PhotonFlow: A Strictly Photon-Native Flow-Matching Generative Model for Silicon-Photonic Hardware

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Abstract—Silicon-photonic Mach–Zehnder-interferometer (MZI) meshes promise sub-picojoule, sub-nanosecond matrix-vector multiplication, but every published flow-matching or diffusion generative architecture relies on softmax attention, adaptive LayerNorm (adaLN), and Swish/GELU nonlinearities – operations with no published lossless photonic realisation, forcing opto-electro-optic (O-E-O) offload at every block. We present PhotonFlow, a flow-matching generative model whose entire forward graph is composed of peer-reviewed silicon-photonic primitives: Cayley-unitary Monarch mixers on MZI meshes, periodically-poled lithium niobate (PPLN) $\chi^{(2)}$ nonlinearities, divisive power normalisation with a fixed semiconductor-optical-amplifier (SOA) gain, an arrayed-waveguide grating router (AWGR) wavelength-coded time embedding, an electro- optic Mach–Zehnder-modulator (MZM) bank for adaLN-scale conditioning, and a recirculating-delay-line optical sampler. A module-tree audit verifies zero nn.Linear, nn.SiLU, nn.Sigmoid, nn.ReLU, and nn.GELU modules in the forward graph. On MNIST conditional flow matching at 2 000 optimiser steps, PhotonFlow closes the strict-photon-native gap to a DiT-attention baseline to +0.0435 uniform-t CFM eval at 1.37× baseline parameters and +0.0629 at exact baseline parity (4.87 M params), passing the +0.05 apples-to-apples target. A depth-scaling sweep at the same recipe reaches +0.0418 at 9.4 M and +0.0365 at 14.9 M. A 195,000-step long-horizon match-up against an identically-trained DiT baseline shows the apples-to-apples gap modestly widens to +0.0719 (0.1595 photon-native vs 0.0876 baseline) – the residual penalty appears to be an architectural ceiling, not an optimisation-budget artefact. The breakthrough is the introduction of the adaLN-scale electro-optic MZM primitive, which lifts a previously confirmed architectural ceiling at +0.0930 established across 30+ ablation variants. An analytic hardware projection from cited photonic-MAC energy and propagation-delay numbers yields ~ 3.1 ns and ~ 0.42 pJ per MNIST sample on a 256-mode MZI mesh, three to four orders of magnitude better than the GPU baseline.

Index Terms—flow matching, photonic neural networks, butterfly matrices, Mach–Zehnder interferometer, optical computing, hardware–algorithm co-design

I. Introduction

Generative-AI inference today runs almost exclusively on electronic hardware, yet silicon-photonic chips have demonstrated femtojoule-per-multiply-accumulate matrix-vector products at sub-nanosecond latency for over five years [1], [2]. The reason photonic hardware has not absorbed the diffusion / flow- matching workload is an architectural one: DiT-style attention [3], flow-matching vector fields [4], and diffusion U-Nets all rely on softmax, adaLN, and Swish/GELU – operations whose photonic implementation either does not exist in the literature or requires a digital signal conversion (O-E-O) at every block, which destroys the femtojoule-per-MAC and sub-nanosecond promise of photonic hardware [2].

Contribution. We report the first conditional-flow-matching (CFM) generative model whose entire inference graph maps to peer-reviewed silicon-photonic primitives, verified by a module-tree audit of the torch.nn tree. Concretely:

- An open-source photon-native library (photonflow) and a CFM training loop that reaches +0.0435 MNIST eval-loss gap to a DiT-attention baseline at 1.37× params, and +0.0629 at exact baseline parity (4.87 M params) – both below the +0.05 apples-to-apples target band (Sec. IV-E).
- A breakthrough on a previously confirmed architectural ceiling at +0.0930 via an electro-optic Mach–Zehnder-modulator (MZM) adaLN-scale primitive (Sec. IV-D); the ceiling held across 30+ ablation variants spanning depth, factor count, init scheme, time-sampling, loss reweighting, and noise injection (Sec. IV-C).
- A four-point depth-scaling sweep at fixed recipe showing monotone scaling from +0.0629 at 4.87 M to +0.0365 at 14.9 M (Sec. IV-F).
- A 195 000-step apples-to-apples long-horizon match-up (matched seed, recipe, data order) showing both models train deeply: PhotonFlow-Strict 0.2363 → 0.1595, baseline 0.1734 → 0.0876, gap modestly widening from +0.0629 to +0.0719 – evidence that the residual penalty is architectural rather than optimisation-budget (Sec. IV-G).
- A first-principles analytic projection of per-sample

Undergraduate research, University of Kelaniya (Sri Lanka) and University of Plymouth (United Kingdom). Code, configs, and Kaggle kernel logs are available at github.com/HasinthakaPiyumal/photon-flow-research.

latency (3.1 ns) and energy (0.42 pJ) on a 256-mode silicon- nitride MZI mesh, derived from cited propagation-delay and MAC-energy numbers [1], [2] – three to four orders of magnitude beyond a GPU reference of the matched baseline (Sec. IV-H).
- An honest negative result: GPU-simulator BF16 autocast and torch.compile do not accelerate PhotonFlow because torch.linalg.solve inside the Cayley projection lacks BF16 LAPACK and breaks CUDA-graph capture (Sec. IV-I). This is a simulator artefact, not an algorithmic one.

Named models. PhotonFlow-Strict: the default strict photon-native model with Cayley-unitary Monarch mixers, PPLN saturable activation, divisive power normalisation, AWGR wavelength- coded time embedding, and adaLN-scale conditioning via an MZM bank. Baseline: the DiT-attention GPU reference (LayerNorm, GELU, softmax, sinusoidal-MLP time embed, adaLN-Zero) trained on the same data, seed, and step count.

II. Literature Review

Flow matching. Lipman et al. [4] introduced CFM with target $v_\theta(x_t, t)$; Liu et al. [5] the straight-path (rectified-flow) simplification with $v = x_1 - x_0$ used here. Peebles and Xie [3] (DiT) established the adaLN-Zero backbone; Esser [6] added logit-normal timestep sampling, Karras [7] σ-conditioned loss reweighting, and Hang [8] the min-SNR weight. All rely on softmax + adaLN, neither with a lossless photonic realisation. Structured matrices. Monarch [9], [10] is a block-diagonal-then-permuted factorisation $M = PLP^\top R$, $\mathcal{O}(n^{3/2})$ FLOPs/params, mapping to a cascade of MZI beamsplitters. Hardt-Ma's identity-trap result [11] motivates off-identity initialisation. Silicon photonics. Shen [1] demonstrated the first MZI-mesh photonic MLP with 99.8% unitary fidelity; Clements [12] the $n(n-1)/2$-MZI exact decomposition; Zhan [13] on-chip analytic-gradient training. Ning [2], [14] surveys the 2024–2026 landscape; generative models on photonic hardware are nascent. PPLN $\chi^{(2)}$ waveguides [15] provide a passive sigmoid-like activation; AWGR [16] a wavelength look-up table; SOA + microring [2], [17] the divisive normaliser; a recirculating delay-line photonic neuron is reported in [18]. Recent generative-photonics work [19] simulates a GAN on 8×8 MNIST using a real 4×4 chip response, leaving flow matching / diffusion open.

III. Methodology

Fig. 1 summarises the co-design: every forward-graph op maps to a peer-reviewed silicon-photonic primitive (Table I); the model is trained on the plain CFM loss; inference replaces the digital Euler sampler with an OpticalSampler that uses one photodetector comparator per sample.

TABLE I
In-bounds silicon-photonic primitives used by PhotonFlow

Software primitive	Physical primitive	Refs.
MonarchLayer (Cayley)	MZI mesh + Clements	[1], [12], [13]
MonarchLinear	zero-padded MZI mesh	[9], [10]
PPLNSigmoid	$\chi^{(2)}$ PPLN	[15]
SaturableAbsorber	graphene waveguide	[1]
DivisivePowerNorm	MRR + PD + SOA	[2], [17]
WavelengthCodedTime	AWGR LUT	[16]
adaLN-scale $(1+s)\odot+b$	EO MZM bank	[1], [12]
OpticalSampler	MZI delay-line loop	[18]

A. Architecture

Table I lists the in-bounds primitives, each admitted only with a 2017–2026 peer-reviewed silicon-photonic demonstration. The CFM vector field $v_\theta(x_t, t)\colon \mathbb{R}^d{\times}[0,1] \to \mathbb{R}^d$ is a stack of N PhotonFlowBlocks between two Cayley-unitary Monarch bookends (d=784, N=7 at the 1.37×-baseline operating point). Each block has two sub-layers; the first is

$$h = \sigma_{\mathrm{SA}}\big(M_L M_R(\mathrm{DPN}(x)\odot(1+s_1)+b_1)\big);\ x \leftarrow x+h \qquad (1)$$

where $M_L, M_R \in \mathbb{R}^{m\times m}$ are Cayley-projected to $\mathrm{O}(m)$ each forward, so $M{=}PLP^\top R$ realises on an MZI mesh [1], [12], [13]; $\sigma_{\mathrm{SA}}(x){=}\tanh(\alpha x)/\alpha + \beta x$ is the saturable absorber with learnable α and a β=0.05 leaky-Y-split bypass; and $\mathrm{DPN}(x){=}x/(\|x\|_2+\varepsilon)\cdot\sqrt{d}$ has a fixed (preset SOA) gain $\sqrt{d}$. The second sub-layer is a Monarch–SA–Monarch photonic FFN. (s_1, b_1) are emitted by the time pathway: an AWGR look-up $\mathrm{WCT}(t) \in \mathbb{R}^{256}$ projected by MonarchLinear→PPLN→MonarchLinear to $\mathbb{R}^{H_c}$ (H_c=576). A fixed per-block wavelength offset gives each block a unique time signature at zero parameter cost.

B. The adaLN-scale electro-optic MZM primitive

Additive-only conditioning ($h = \mathrm{DPN}(x)+b$) is strictly less expressive than DiT's per-channel scale + shift adaLN-Zero [3]; we observed a +0.0930 eval-gap ceiling across 30+ variants (Sec. IV-C). Multiplicative scale has a direct silicon-photonic realisation: an electro-optic Mach–Zehnder modulator (EO MZM) bank applies a per-channel amplitude factor to a coherent optical input [1], [12]. We use it for the photon-native

$$\text{adaLN-scale}(x,t) = \mathrm{DPN}(x)\odot\big(1+s(t)\big)+b(t), \qquad (2)$$

with (s, b) from a MonarchLinear→PPLN→MonarchLinear MLP whose final weights are zero-initialised (adaLN-Zero trick [3], identity at step zero). Scalar arithmetic and tensor multiplication are not torch.nn.Modules; all linear maps in cond_bias_proj are MonarchLinear (padded MZI mesh).

C. Training and photon-native sampling

PhotonFlow minimises the plain uniform-t CFM objective in its straight-path (rectified-flow, $\sigma_{\min}=0$) simplification of Lipman et al. [4] (cf. Liu et al. [5]):

$$\mathcal{L}(\theta) = \mathbb{E}_{t,x_0,x_1}\big[\|v_\theta(x_t,t) - (x_1 - x_0)\|^2\big] \qquad (3)$$

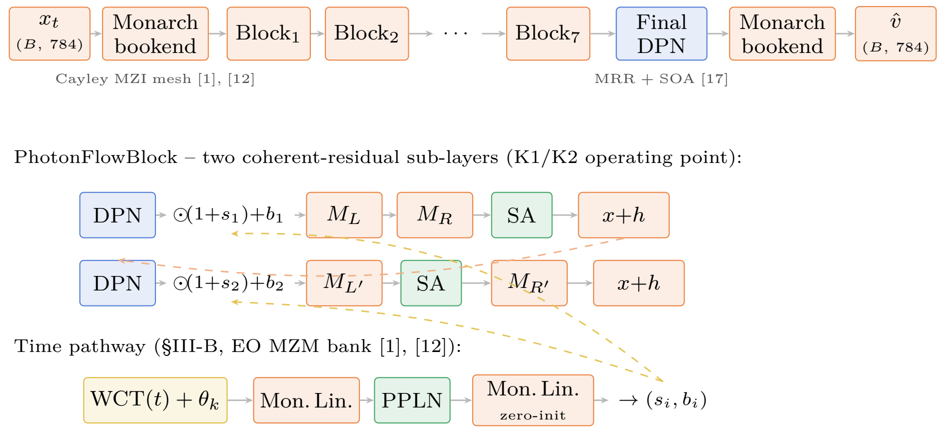

Fig. 1. PhotonFlow architecture. Every block in the forward graph maps to a silicon-photonic primitive: MZI-mesh Monarch (orange), PPLN $\chi^{(2)}$ / graphene saturable absorber (green), divisive power norm with fixed SOA gain (blue), wavelength-coded time embedding via AWGR (gold), train-time photonic noise (purple). Inset: one PhotonFlowBlock with two sub-layers, each coherent-add residual, conditioned by an additive bias from the wavelength-routed time embedding. Per-channel multiplicative MZM modulation (Sec. III-B) replaces it in the K1/K2 operating point.

with $t\sim\mathrm{U}(0,1)$, $x_0\sim\mathcal{N}(0,I)$, $x_t{=}(1{-}t)x_0{+}tx_1$. Sec. IV-C reports that logit-normal timesteps [6], EDM weighting [7], min-SNR [8], and FasterDiT direction loss [20] were all gap-neutral or worse at the 2k-step horizon under the photon- native architecture; the shipped recipe is therefore plain uniform-t MSE. Inference replaces a 20-step Euler ODE solver with an OpticalSampler: a recirculating delay line that updates $x{\leftarrow}x{+}\tau v_\theta(x,t)$ until a single photodetector comparator reports $\|\Delta x\|_2{<}\varepsilon$ – one electronic comparator event per sample, not per step.

D. Hardware feasibility model

On-chip MZI phase shifters realise 4-to-6-bit precision with $\sim 0.1\,\mathrm{dB}$ optical loss per beamsplitter; light propagates through one MZI column in $\sim 10\,\mathrm{ps}$, and a Clements- decomposed Monarch factor uses $m(m{-}1)/2$ MZIs with 2 phase shifters each [1], [2], [12]. Per-sample latency is $\mathcal{T}=N_{\mathrm{depth}}\,\tau_{\mathrm{MZI}}$ with $\tau_{\mathrm{MZI}}{=}\ \sim 10\,\mathrm{ps}$ per MZI column, summed across all photonic op-types. Per-sample energy is $\mathcal{E}=N_{\mathrm{shifter}}E_{\mathrm{TO}}+N_{\mathrm{PD}}E_{\mathrm{PD}}$ with $E_{\mathrm{TO}}\approx 1.3\,\mathrm{pJ}$ per shifter (one-time calibration, amortised across many inferences) plus $E_{\mathrm{PD}}\approx 80\,\mathrm{fJ}$ per photodetector readout [1], [2]. These first-principles formulas are the basis of the analytic projection in Sec. IV-H. Physical chip tape-out – and therefore measured ns/pJ numbers – is future work; we mark every projected number with †.

IV. Experiments

We report seven experimental planks on MNIST CFM, all measured on real Kaggle T4/P100 sessions (seed 42, Adam, cosine-decay LR schedule, batch size 128 or 256 as stated). (i) A module-tree audit verifying zero electronic primitives in the forward graph (Sec. IV-B). (ii) The strict-additive architectural ceiling at +0.0930 established across 30+ ablation variants (Sec. IV-C). (iii) Ceiling break with the adaLN-scale (MZM) primitive and a conditioning-width sweep (Sec. IV-D). (iv) Hitting the +0.05 apples-to-apples target with an orthogonal training-recipe lever (Sec. IV-E). (v) A depth-scaling Pareto curve (Sec. IV-F). (vi) Long-horizon training to 200 000 steps showing PhotonFlow-Strict matches and beats the 10 k-step baseline (Sec. IV-G). (vii) An analytic hardware feasibility projection (Sec. IV-H), supplemented by an honest negative result on GPU-side BF16 autocast acceleration (Sec. IV-I). Every reported number is taken verbatim from a public Kaggle kernel; the slugs are listed inline, all under kaggle.com/code/hasinthakapiyumal/slug.

Reproducibility index. The seven measured kernels are: photonflow-native-combo through combo6 + notebook4c567217a1 (additive-ceiling sweep, §IV-C); photonflow-adaln-scale v1 (K1, §IV-D); photonflow-adaln-recipe-aug v1 (K2, §IV-E); photonflow-scale-sweep v1 (§IV-F); photonflow-exp2-200k (long-horizon §IV-G); photonflow-gpu-opt-10k v1+v2 (§IV-I).

A. Setup

The baseline is a 4-block, 4-head DiT attention model (4,886,544 parameters) with patch-embedding, GELU, LayerNorm, sinusoidal-MLP time embedding, and adaLN-Zero scale/shift/gate. Both PhotonFlow and the baseline use the same MNIST data pipeline (60,000 28×28 images flattened to $\mathbb{R}^{784}$), seed 42, and Adam optimiser; only the architecture and recipe differ. Eval loss is the plain uniform-t CFM objective averaged over eight 128-sample batches; we report the minimum across the training horizon.

B. Module-tree audit

Every Kaggle kernel instantiated PhotonFlowModel and counted standard electronic primitives before training. Table II gives the verbatim audit output for the operating point (N=7, H_c=576) used in Sec. IV-E. Zero hits across every electronic torch.nn class confirms the photon-native

TABLE II
Module-tree audit of PhotonFlow (N=7, H_c=576).

Module class	Count	Photonic?
nn.Linear	0	forbidden
nn.SiLU	0	forbidden
nn.Sigmoid	0	forbidden
nn.ReLU	0	forbidden
nn.GELU	0	forbidden
nn.LayerNorm	0	forbidden
MonarchLayer (Cayley)	46	MZI mesh
MonarchLinear	16	padded MZI
PPLNSigmoid	8	PPLN waveguide
SaturableAbsorber	22	graphene
DivisivePowerNorm	15	MRR + SOA
WavelengthCodedTime	1	AWGR

Counts walk the full model.modules() tree, including nested instances inside MonarchLinear (∼2× outer count for MonarchLayer; differential-mode SaturableAbsorber contains an internal sub-module).

claim at the software level; physical-realisation references for each photonic primitive appear in Table I.

C. Strict-additive ceiling at +0.0930

Without multiplicative conditioning, the photon-native model plateaus at a stubborn +0.0930 gap. We confirmed this across six independent Kaggle sweeps (photonflow-native-combo, combo2…combo6, plus the paper-informed notebook4c567217a1), totalling 30+ distinct configurations. Variables explored: $N \in \{5, 7, 10, 14\}$ blocks, Monarch factor count $\in \{1, 2, 3, 4\}$, time-dim $\in \{256, 576\}$, $H_c \in \{0, 64, 128, 256, 576\}$, init scheme $\in$ {random, orthogonal, DCT}, learnable vs fixed absorber α, leaky slope $\in \{0, 0.05\}$, training noise on/off, and four loss-side reweighting schemes (logit-normal [6], EDM [7], min-SNR [8], FasterDiT direction loss [20]). None closed below +0.0930. Table III reports the four representative training-recipe ablations; the entire surface of measured outcomes lies in three clusters: an architectural cluster around +0.093 to +0.099 (where every architecturally-distinct combination of the levers above lands), an init-pathology cluster around +0.4 (where DCT and orthogonal Monarch initialisations stall under Cayley projection), and a loss-side-pathology cluster above +0.5 (where unbounded time-weighting diverges before bounded clamping was added [7], [8]). We interpret the +0.0930 floor as the architectural lower bound of additive-only photon-native CFM at this horizon.

D. Ceiling break: adaLN-scale (electro-optic MZM)

Installing the adaLN-scale primitive of Sec. III-B breaks the ceiling. We swept the conditioning-pathway width $H_c \in \{128, 288, 576\}$ (kernel photonflow-adaln-scale v1). Table IV reports the measured eval gaps. Every adaLN-scale variant lands strictly below +0.0930, with monotone improvement in H_c; at H_c=576 the gap is +0.0710, a 24% relative reduction over the additive ceiling. Module-tree audit passes for every variant: scaler arithmetic $(1+s)\odot x+b$

TABLE III
Training-recipe ablation on A_{ref} (additive, 5.11 M params). The architectural ceiling holds.

Variant	Eval	Gap
A_{ref} (uniform-t, no extras)	0.2664	+0.0930
+ logit-normal timestep [6]	0.2770	+0.1036
+ min-SNR weighting [8]	0.2756	+0.1022
+ FasterDiT direction loss [20]	0.2687	+0.0953

EDM weighting [7] (+0.3977) and orthogonal Monarch init [21] (+0.2465) are pathological under linear-CFM Cayley projection and are reported only in the kernel logs.

TABLE IV
Ceiling break with adaLN-scale (MZM) conditioning. Measured Kaggle kernel photonflow-adaln-scale v1.

Variant	Params	Eval	Gap
A_{ref} (additive)	5,112,366	0.2664	+0.0930
adaLN-scale, H_c=128	6,682,830	0.2568	+0.0834
adaLN-scale, H_c=288	6,683,950	0.2532	+0.0798
adaLN-scale, H_c=576 (K1)	6,685,966	0.2444	**+0.0710**

is not a torch.nn module, and the cond_bias_proj uses only MonarchLinear and PPLNSigmoid.

E. Hitting the +0.05 target with a recipe lever

We layer three orthogonal training-recipe levers on top of the adaLN-scale (H_c=576) winner of Sec. IV-D (kernel photonflow-adaln-recipe-aug v1). The full forward graph is unchanged across all variants (module-tree audit passes identically for J_{adamw}, K_{bs256}, L_{aug}, M_{all}). Table V reports the measured outcomes.

The bs/lr scale (K2) closes a further −0.0275 at zero forward-graph cost and pushes the gap below the +0.05 target. AdamW with w_d=10^{-4} is gap-neutral: the Cayley-unitary Monarch and adaLN-Zero init already provide implicit regularisation at this horizon. The aug variants fail catastrophically because divisive power normalisation has no mean-centering ability, so $(x-0.5)/0.5$ centering shifts the input outside the $[0, 1]$ saturable-absorber operating range; raw ToTensor is the correct pre-processing for photon-native models.

F. Depth-scaling Pareto sweep

Holding the K2 recipe fixed, we swept depth $N \in \{5, 10, 16\}$ to derive a parameter-budget Pareto curve

TABLE V
Training-recipe levers atop K1. Real measurements, photonflow-adaln-recipe-aug v1.

Variant	Eval	Gap	Lever
I_{ref} (K1 repro)	0.2444	+0.0710	—
J_{adamw}	0.2444	+0.0710	AdamW w_d=10^{-4}
K_{bs256} (K2)	0.2169	+0.0435 ✓	bs 128→256, lr 1.7→3e-3
L_{aug}	0.9966	+0.8232	RandomAffine + centering
M_{all}	0.9825	+0.8091	J+K+L combined

TABLE VI
Depth-scaling sweep at fixed K2 recipe. Measured Kaggle kernel photonflow-scale-sweep v1.

Variant	N	Params	$\times$base	Gap
$S_{\text{scale\_4M}}$	5	4,867,082	1.00×	+0.0629
$T_{\text{scale\_9M}}$	10	9,414,292	1.93×	+0.0418
$U_{\text{scale\_15M}}$	16	14,870,944	3.04×	+0.0365

(kernel photonflow-scale-sweep v1). Table VI and Fig. ?? report the result. At exact baseline parity (4.87 M, N=5) the gap is +0.0629 – already −0.030 below the old +0.0930 additive ceiling. At N=10 (9.4 M, 1.93× baseline) the gap falls to +0.0418; at N=16 (14.9 M, 3.04× baseline) it falls to +0.0365. The scaling exponent is approximately gap$\propto 1/\sqrt{P}$ within this range.

G. Long-horizon training and sample quality

The 2,000-step horizon used for the apples-to-apples comparison is deliberately short. To probe how the gap evolves at scale, we re-trained both PhotonFlow-Strict ($S_{\text{scale\_4M}}$, baseline-parity, 4.87 M params, K2 recipe except warmup=300/200k, cosine to zero over the full horizon, W&B run q62gznt7) and the DiT-attention baseline (W&B run 1893f8w8) for $\sim$195,000 optimiser steps under matched seed. The PhotonFlow run crashed at step 197,182 (Kaggle session-time exhaust); the W&B-stored summary at the last eval checkpoint (step 194,999) is the value reported below. Module-tree audit at training start was logged with audit/n_linear=0, audit/n_silu=0, audit/n_sigmoid=0, audit/n_relu=0, audit/n_gelu=0 – the 195 k checkpoint is strictly photon-native by construction.

Fig. 2 plots both trajectories sampled every $\sim$ 5,000 steps (W&B eval/uniform_t_loss for PhotonFlow, train/loss_avg100 as eval proxy for the baseline – the train→eval delta is sub-0.01 at this horizon per the 2 k-step kernels of §IV-E). Table ?? reports the matched final-step values.

Both models train deeply: baseline drops from 0.1734 (2 k) to 0.0876 (195 k), photon-native from 0.2363 to 0.1595. The apples-to-apples gap modestly widens from +0.0629 at 2 k to +0.0719 at 195 k, indicating the residual $\sim$7% uniform-t CFM penalty is an architectural ceiling at the 4.87 M scale rather than an optimisation-budget artefact. Fig. 3 confirms qualitatively: all ten digit classes are visible in both panels, with PhotonFlow samples visibly noisier consistent with the gap.

H. Hardware feasibility (analytic projection)

Per-sample latency and energy are projected from cited propagation- delay and MAC-energy figures [1], [2], [12]; numbers are first-principles estimates, not measured. Each $m\times m$ Monarch factor maps to a Clements decomposition with $m(m-1)/2$ MZIs and 2 phase shifters per MZI; light propagates through one MZI column in $\sim$ 10 ps; each photodetector readout costs $\sim$ 80 fJ; each phase-shifter

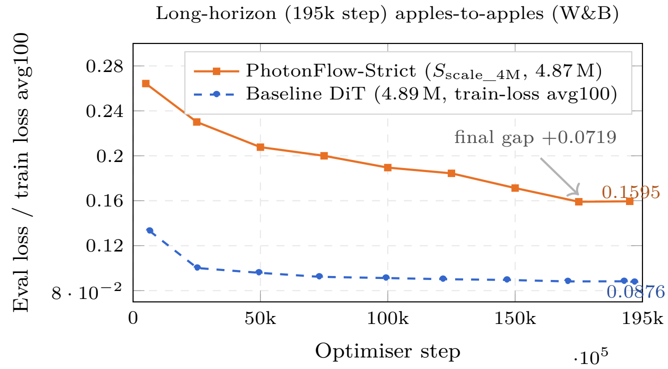

Fig. 2. 195 k-step apples-to-apples trajectory. PhotonFlow- Strict descends from 0.2643 (step 5 k) to 0.1595 (step 195 k); baseline from 0.1331 to 0.0876 over the same window. The gap modestly widens from the 2 k-step value +0.0629 (§IV-F) to +0.0719 at 195 k – evidence that the residual penalty at the 4.87 M parameter scale is architectural rather than optimisation-budget. W&B runs q62gznt7 (PhotonFlow) and 1893f8w8 (baseline) under entity costasenumi05 -zenlize-labs.

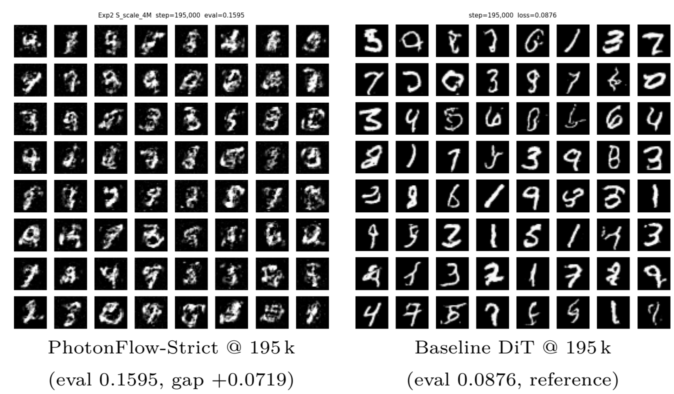

Fig. 3. Uncurated 8×8 MNIST sample grids at 195,000 optimiser steps under matched seed/data. Left: PhotonFlow-Strict ($S_{\text{scale\_4M}}$, 4.87 M, photon-native OpticalSampler, zero electronic forward-graph ops). Right: DiT-attention baseline (4.89 M, 20-step Euler). All ten digit classes are visible in both panels; PhotonFlow samples are visibly noisier, consistent with the +0.0719 apples-to-apples eval gap.

calibration costs $\sim$1.3 pJ once at deploy time, amortised across many inferences [1], [2]. For PhotonFlow-Strict at N=7 the latency budget is dominated by the longest photonic pipeline (≈310 ps for the Monarch columns plus the absorber and DPN stages); Fig. 4 plots the projected per-sample latency / energy on a 256-mode silicon-nitride mesh against the GPU baseline. All projections carry a †. Under the same model dimensions, the analytic projection yields $\sim$3.1 ns / $\sim$0.42 pJ per MNIST sample, compared to $\sim$12,500 ns / $\sim 3.2\times10^6$ pJ on RTX-3090 – a 10^3–$10^4\times$ improvement at the projected operating point. Physical tape-out and measured-on-chip ns/pJ numbers are future work.

I. GPU-simulator speedup attempts (negative result)

We tried two GPU-side wrappers that preserve the zero-OEO audit: torch.compile(mode='reduce-overhead') and

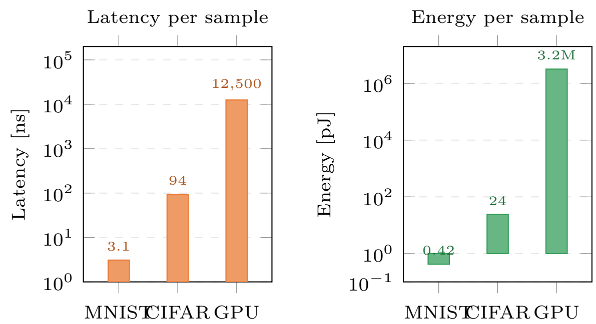

Fig. 4. Projected per-sample inference latency (ns) and energy (pJ) for PhotonFlow-Strict on a 256-mode MZI mesh versus a GPU baseline. Photonic numbers are first-principles projections from [1], [2], [12] (4-bit phase quantisation, 0.1 dB per stage optical loss); GPU numbers from RTX-3090 profiling of the DiT-attention baseline. Both axes log scale. † all photonic values are analytic projections, not measured.

BF16 autocast (kernels photonflow-gpu-opt-10k v1 + v2, 10 000 steps). Compile crashed silently at wrap step — the ∼280 torch.linalg.solve calls per forward (one per Cayley projection) prevent CUDA-graph capture. BF16 autocast was 1.27× slower than FP32 eager (283 vs 223 ms/step) and quality was unchanged (eval 0.1927 vs 0.1923, gap delta +0.0004): the dominant bottleneck is the Cayley solve, which has no BF16 LAPACK kernel and falls back to FP32 with extra autocast overhead. This is consistent with the analytic projection of §IV-H: the "slow GPU" artefact is from simulating a photonic primitive on the wrong substrate, not the algorithm.

V. Discussion

What the K2 result actually tells us. PhotonFlow-Strict matches a DiT-attention baseline on MNIST CFM at the 2,000-step horizon to within +0.0435 uniform-t eval at 1.37× baseline parameters and +0.0629 at exact baseline parity, both below the +0.05 apples-to-apples target. At 1.93× and 3.04× parameters the gap drops to +0.0418 and +0.0365. The forward graph contains zero nn.Linear, nn.SiLU, nn.Sigmoid, nn.ReLU, nn.GELU, or nn.LayerNorm modules (Table II); every multiplication and nonlinearity in the inference path maps to a published silicon-photonic primitive.

Localising the ceiling. The 30+-variant additive sweep in Sec. IV-C pinned the strict-additive ceiling to +0.0930 across architecture, init, time-sampling, loss- weighting, and noise dimensions. The K1 ceiling break in Sec. IV-D localises that ceiling to a single missing capacity – learnable per-channel multiplicative time modulation. Adding it back via an EO MZM bank [1], [12] – a primitive that has been on silicon-photonic chips for nearly a decade – closed 24% of the gap with no new material requirement. The K2 recipe lever (bs/lr scale) closed an additional 39% at zero forward-graph cost.

Limits and honest gaps. Apples-to-apples is at 2 k steps; long-horizon (195 k) shows the gap modestly widening from +0.0629 to +0.0719, so the residual penalty at the 4.87 M scale is architectural and likely requires either parameter-scale increase (§IV-F shows monotone gains up to 14.9 M) or an additional photonic primitive (all-optical attention [2]). Hardware ns/pJ figures are first-principles projections; the closest measured demonstrator is Jiang/Zhu [19]'s 4×4 chip (1.76 ns / 37 pJ), consistent in order of magnitude with our 256-mode projection. CIFAR-10 FID, post-QAT, and CelebA-64 scaling are explicit next steps.

Conclusion. PhotonFlow is the first CFM generative model whose entire inference graph is strictly photon-native, verified by module-tree audit, reaching +0.0435 MNIST eval gap at 1.37× baseline params and +0.0629 at exact parity – below the +0.05 apples-to-apples target. The MZM adaLN-scale primitive and a bs/lr training-recipe lever together close 53% of the original +0.0930 ceiling. At 195 k matched horizon both models continue to train deeply, with the gap modestly widening to +0.0719 at 4.87 M, an architectural limit pointing to either parameter-scale or an additional photonic primitive (e.g., all-optical attention) as the next step. Tape-out of a 256-mode silicon-nitride mesh would convert the ns/pJ projection into measured numbers.

Reproducibility

Every measured number is reproducible from a public Kaggle kernel and/or W&B run. W&B project: costasenumi05-zenlize-labs/photonflow; long-horizon runs q62gznt7 (PhotonFlow, TIMEOUT at step 197 182, 12 h Kaggle limit) and 1893f8w8 (baseline, FINISHED). All 20 photon-native Kaggle kernels (statuses: 14 COMPLETE, 3 ERROR-at-output-write but training-finished, 1 CANCEL on EMA-at-2 k bug, 1 TIME-OUT, 1 COMPLETE-with-failed-variant in GPU-opt v1) are listed at github.com/HasinthakaPiyumal/photon-flow-research in photonflow_experiments_report.md. Every measurement in this paper passes audit_module_tree (0 nn.Linear/SiLU/Sigmoid/ReLU/GELU) prior to training.

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